

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.0000000

Multi-Domain Vision Transformer Fusion for Intersectional Demographic Classification from Facial Images

RICKY EKA PUTRA¹, REZKY ARISANTI PUTRI¹, YUNI YAMASARI¹, RAFY AULIA AKBAR¹,
AND SHUKOR SANIM MOHD FAUZI²

¹Department of Informatics, Faculty of Informatics, Universitas Negeri Surabaya, Surabaya 60231, Indonesia

²Faculty of Computer and Mathematical Sciences, Universiti Teknologi MARA Perlis Branch, 02600 Arau, Perlis, Malaysia

Corresponding author: Ricky Eka Putra (e-mail: rickyeka@unesa.ac.id).

This research was supported in part by the Department of Informatics, Faculty of Informatics, Universitas Negeri Surabaya.

ABSTRACT Simultaneous recognition of facial demographic attributes such as race and gender faces challenges from expression variations, biological aging, phenotypic overlap, and the limitations of single-domain representations. Therefore, a unified representation framework capable of comprehensively integrating diverse facial characteristics is required. This study proposes a multi-domain feature fusion framework that integrates visual representations from three task-specific pre-trained Vision Transformer (ViT) models for facial biometrics, affective expression, and age estimation, coupled with classical machine learning pipeline optimization for classifying six intersectional demographic subgroups on the DemogPairs dataset. Visual representations are extracted offline from each model to produce a unified visual representation that combines complementary cross-domain information. Seven feature configurations are evaluated through 5-Fold Stratified Cross-Validation across four classifiers: Random Forest (RF), Gaussian Naive Bayes (GNB), Logistic Regression (LR), and Support Vector Machine (SVM) with hyperparameter tuning via Grid Search Cross-Validation (GridSearchCV). Experimental results demonstrate that tri-domain fusion achieves the best performance across three of the four evaluated classifiers, with the SVM model yielding the highest performance, recording an Accuracy of 93.70%, Precision of 93.72%, Recall of 93.70%, and F1-Score of 93.69% on independent test data. In-depth evaluation across the six subgroups records an F1-Score range between 91.74% and 96.14%, demonstrating the effectiveness of multi-domain representation fusion in recognizing intersectional demographic attributes.

INDEX TERMS Race and gender classification, intersectional demographic recognition, multi-domain feature fusion, Vision Transformer.

I. INTRODUCTION

Automated facial demographic attribute recognition plays a pivotal role across various intelligent application domains, including digital forensics systems, biometric access control, human-computer interaction, and personalized interactive services [1], [2]. However, multiple studies have reported performance disparities across specific demographic subgroups, including females and individuals with darker skin tones [3], [4]. Such disparities can be influenced by various factors, including data distribution imbalances and the limitations of feature representations in handling visual variations [5]. The limitations of conventional systems in coping with such visual variations demonstrate that fair and accurate

demographic recognition still faces tangible practical challenges, thereby necessitating novel representation approaches capable of modeling the interaction between racial and gender demographic attributes in an integrated and comprehensive manner.

While some facial attribute recognition approaches model attributes separately, multi-task learning enables multiple attributes to be learned simultaneously [6]. Modeling attributes in isolation can restrict the analysis of groups formed by attribute combinations, such as race and gender [7], [8]. Consequently, aggregate metrics potentially fail to capture performance disparities across specific subgroups [9]. Evaluation across demographic subgroups with controlled, bal-

anced representation can provide a more consistent and objective comparative foundation for measuring inter-group performance variations [9]. Intersectional modeling also faces representational challenges because facial characteristics and expressions can vary across demographic groups, thereby necessitating visual representations capable of preserving discriminative information while remaining robust against such variations [10].

The methodological evolution of facial attribute recognition has shifted from handcrafted features toward deep learning paradigms, including CNN and ViT architectures. In the era of Convolutional Neural Networks (CNN), various studies explored demographic feature extraction by leveraging specific facial regions such as the mid-face area, multi-scale multi-task frameworks for simultaneous age and gender estimation, and deep CNN models for ethnicity recognition [11], [12], [13]. ViT is capable of modeling global inter-patch relationships via Multi-Head Self-Attention (MHSA), thereby potentially yielding more contextual visual representations, while the utilization of synthetic generative representations has also been developed to mitigate demographic disparities [14], [15]. In addition, the integration of parallel attention mechanisms, the utilization of ViT for facial gender classification, and the exploration of multi-axis transformer architectures have further enriched the diversity of visual representation modeling [16], [17], [18]. Furthermore, exploration into combining cross-domain ViT representations has begun to emerge through the integration of dual facial and emotional features to enhance race and gender attribute discrimination [19], [20]. Nevertheless, the fusion of biometric, expression, and aging representations into a unified representation remains relatively limited.

Despite continuous advancements in ViT, several research gaps directly relevant to intersectional demographic classification remain. First, the majority of prior studies rely on single-domain facial biometric representations, while the simultaneous integration of affective information for intersectional demographic classification remains limited [11], [17], [18]. Second, demographic recognition frequently exhibits high sensitivity to age-related facial morphological changes because the employed feature extraction does not incorporate biological aging representations [12], [17]. Third, comparative studies regarding the influence of classifiers, feature scaling, and dimensionality reduction on transformer representations remain limited [13], [19], [20]. To bridge the limitations of single-domain representations and evaluate these downstream classifier decision boundaries, this study proposes an integration of three ViT representation domains combined with structured classical classifier pipeline optimization.

To address these research gaps, this study presents a unified multi-domain representation fusion evaluation framework that combines three task-specific representations for intersectional demographic classification on facial images. This evaluation framework integrates three complementary representation domains, encompassing facial biometric representations, affective emotional expressions, and biological

aging characteristics, into a single unified representation via feature fusion [21], [22]. Feature extraction is conducted offline from frozen pre-trained ViT backbone models, allowing downstream classification to be executed efficiently without requiring the computational overhead of backbone fine-tuning [23]. The resulting fused feature vectors are subsequently evaluated systematically through single-domain, dual-domain, and tri-domain ablation schemes using four families of classical machine learning classifiers, namely RF, GNB, LR, and SVM. All downstream classifiers are optimized and evaluated using Stratified Cross-Validation within a structured learning pipeline that integrates feature scaling and Principal Component Analysis (PCA) dimensionality reduction, with each transformation strictly learned only on the training data split of each fold to prevent data leakage [24]. This approach enables an objective empirical analysis regarding the relative contribution of each representation domain to the robustness of demographic decision boundary modeling.

The primary scientific contributions of this study are summarized into three key points as follows:

- 1) Formulating a multi-domain representation fusion evaluation framework that integrates facial biometric, affective expression, and aging estimation domains from three frozen pre-trained Vision Transformer models, combined with classical machine learning pipeline optimization (SVM, LR, RF, GNB) and systematic hyperparameter tuning via 5-Fold Stratified Cross-Validation.
- 2) Presenting a comprehensive comparative evaluation across 28 model configurations on the DemogPairs dataset, confirming the superiority of tri-domain fusion on the nonlinear SVM model, and inferentially demonstrating the occurrence of feature representation saturation and classifier-dependent fusion dynamics.
- 3) Providing a granular investigation of performance disparities across six intersectional demographic subgroups through formal fairness metrics, evaluation of within-race and cross-gender error patterns, and positioning as the first comprehensive controlled 6-class baseline on the DemogPairs dataset.

The remainder of this article is organized as follows: Section II discusses prior research on facial demographic recognition and transformer architectures. Section III details the DemogPairs dataset specifications, multi-domain ViT representation extraction, and the optimized classification pipeline. Section IV presents the experimental evaluation results, feature ablation studies, and intersectional subgroup performance analysis. Finally, Section V summarizes the primary research conclusions, methodological limitations, and recommendations for future research directions.

II. RELATED WORKS

Facial demographic attribute recognition has transitioned from region-based descriptors and handcrafted features toward deep representations. Local region-based approaches leverage the geometry of the central facial region to recognize

ethnicity and mitigate background interference [11]. However, CNN-based approaches still face challenges in maintaining performance under illumination variations and facial characteristic differences across racial groups, including the Other-Race Effect phenomenon. Increasing the depth of convolutional networks can enrich visual feature representations to differentiate diverse populations [13]. In addition, parallel feature-sharing mechanisms with visual attention have been developed to capture inter-attribute correlations while mitigating the limitations of local representations. This approach reflects a shift from feature engineering toward representation learning that is more adaptive to facial diversity [16].

The application of ViT in facial biometrics offers visual modeling capabilities distinct from convolutional approaches. Through the Multi-Head Self-Attention (MHSA) mechanism, transformers can model inter-patch relationships globally, thereby capturing broader spatial dependencies compared to local convolution operations [14]. These characteristics demonstrate the potential of transformers in modeling demographic attributes, including gender, on standardized-resolution facial images [17]. Beyond standard ViT architectures, hybrid architectures such as MaxViT combine window-based and grid-based attention to construct hierarchical spatial representations [18]. Although transformers possess powerful global representation capabilities, their performance remains influenced by the quality, size, and distributional characteristics of the training data utilized during the learning process.

Investigations into performance disparities in computer vision systems have driven evaluation protocols that account for balanced class and demographic subgroup distributions. Testing on balanced datasets provides controlled experimental conditions for comparing performance across subgroups by mitigating the confounding impact of sample size imbalances. However, distributional balance does not inherently eliminate social biases embedded within the data. Analysis of learned representations also demonstrates that transformer models can reproduce social biases originating from large-scale pre-training corpora [14]. To reduce performance disparities across racial and gender groups, generative augmentation based on StyleGAN2 has been employed to synthesize variations that help mitigate classification errors [15]. These findings underscore the critical importance of systematically conducting evaluations across demographic subgroups.

The limitations of single-domain representations in capturing the diversity of facial characteristics have motivated the exploration of multi-task learning and representation fusion from multiple information sources. Multi-task approaches have been utilized to learn correlations between age and gender attributes through jointly processed facial representations [12]. Parallel representation-sharing mechanisms have also been developed to combine diverse affective attributes simultaneously, whereas generative approaches can enrich feature variations across demographic groups [16], [15]. Furthermore, the integration of ViT-based feature representations uniting facial biometric cues and emotional expressions has

demonstrated potential for improving intersectional attribute classification [19]. Combining biometric representations with age information also exhibits complementary informational contributions, establishing cross-domain fusion as a highly relevant approach [20].

Once feature representations are obtained, the choice of downstream classifier governs how high-dimensional features are mapped to the decision space. End-to-end approaches commonly employ a SoftMax layer to map feature representations directly to class probabilities [11], [18]. Alternatively, Support Vector Machines (SVM) can learn non-linear decision boundaries within the feature space without updating the frozen backbone parameters [13]. Tree-based ensemble models such as RF and probabilistic approaches such as GNB can also serve as classifiers with distinct complexity and class-separation characteristics [12]. These differences demonstrate that classifier selection substantially influences the effectiveness of feature representation utilization, particularly when features originate from multiple domains and exhibit high dimensionality.

Based on the reviewed literature, significant opportunities remain to explore facial demographic attribute modeling through the integration of representations across multiple domains. In contrast to prior studies that predominantly rely on single-domain representations or partial domain combinations, this study proposes a tri-domain fusion framework that unifies facial biometric geometric representations, affective expression cues, and biological aging morphological characteristics using pre-trained ViT models with frozen backbones. The deep feature representations from the three domains are subsequently concatenated and fed into an SVM to learn non-linear decision boundaries. Evaluations are conducted on the balanced DemogPairs dataset to benchmark intersectional race and gender classification while systematically analyzing performance disparities across all evaluated subgroups.

III. MATERIALS AND METHODS

The proposed framework integrates multi-domain visual representations and machine learning model optimization for intersectional race and gender classification, as illustrated in Figure 1. The methodology begins with partitioning 224×224 pixel resolution facial images from the DemogPairs dataset into training and test partitions via a stratified split. Three frozen pre-trained ViT-Base models extract 768-dimensional visual representations from the [CLS] token offline. The three vectors are combined through concatenation fusion $\mathbf{z}_{\text{tri}} = \mathbf{f}_{\text{face}} \oplus \mathbf{f}_{\text{emotion}} \oplus \mathbf{f}_{\text{age}}$, where $\mathbf{z}_{\text{tri}}$ denotes the 2,304-dimensional tri-domain vector, $\oplus$ is the concatenation operator, and $\mathbf{f}_{\text{face}}$, $\mathbf{f}_{\text{emotion}}$, and $\mathbf{f}_{\text{age}}$ represent facial biometric, affective expression, and age features, respectively. The representation vectors are processed within a modular Scaler and PCA pipeline before being classified into six intersectional classes by RF, GNB, LR, and SVM. Hyperparameter tuning is conducted via 5-Fold Stratified Cross-Validation with an isolated Grid-SearchCV scheme to prevent information leakage prior to test data evaluation.

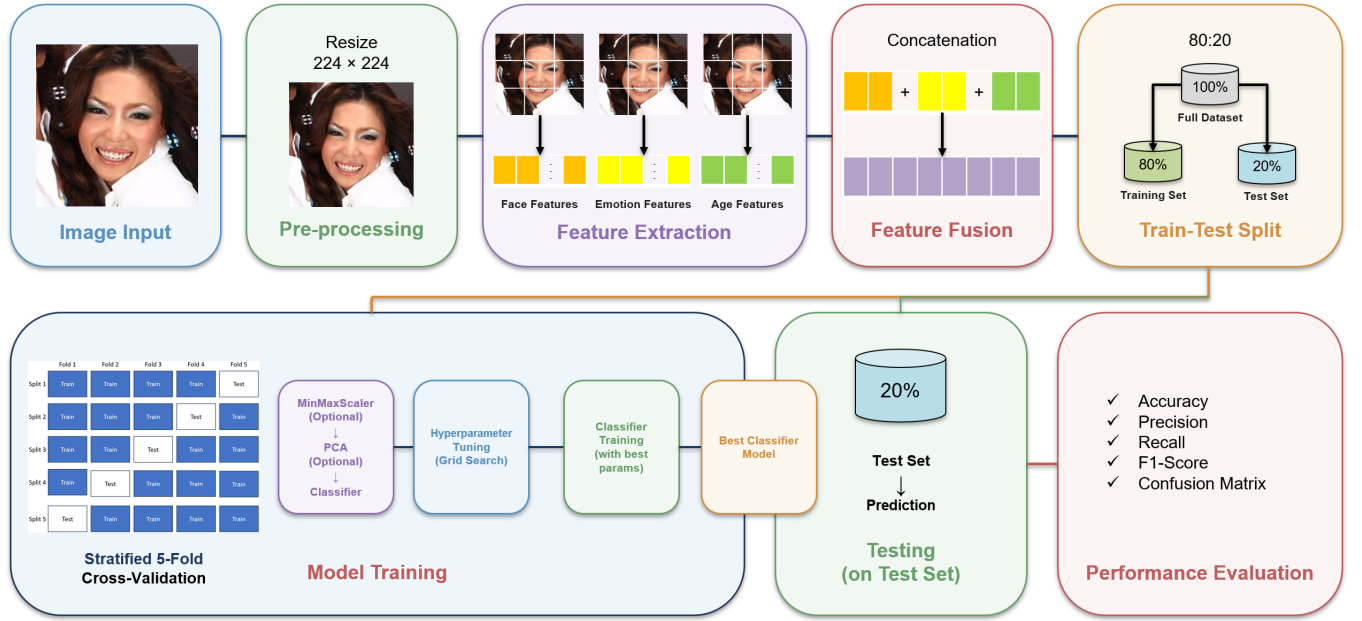


FIGURE 1. End-to-End Framework of the Proposed Multi-Domain Vision Transformer Feature Fusion and Classical Classifier Optimization for Intersectional Race and Gender Classification.

A. DATASET

The dataset used in this study is DemogPairs [25], a dataset specifically designed for facial recognition across demographic groups. This dataset contains a total of 10,800 facial images distributed evenly across six intersectional classes. The six subgroups encompass cross-combinations of three macro-race groups and two gender groups, namely Asian Females, Asian Males, Black Females, Black Males, White Females, and White Males, with a uniform allocation of exactly 1,800 images per class. This balanced data structure provides a controlled evaluation condition to objectively compare model performance across subgroups, while simultaneously mitigating distortions arising from sample size disparities [20]. The visual characteristics of the six demographic subgroups are illustrated in Figure 2.

TABLE 1. Dataset Partition and Demographic Subgroup Distribution.

Subgroup	Train Set (80%)	Test Set (20%)	Total
Black_Males	1,440	360	1,800
White_Females	1,440	360	1,800
Asian_Males	1,440	360	1,800
White_Males	1,440	360	1,800
Black_Females	1,440	360	1,800
Asian_Females	1,440	360	1,800
Total	8,640	2,160	10,800

To ensure the integrity of empirical testing, the dataset is partitioned using an 80/20 stratified split protocol with random_state=42 and parameter stratify=y. This partition yields 8,640 training images with 1,440 samples per class and 2,160 independent held-out test images with 360 samples per class across the six intersectional classes, as detailed in Table 1.

This partitioning completely isolates the held-out test set, ensuring that all test data remain strictly unseen during hyperparameter tuning as well as PCA and Scaler fitting. In the image standardization stage, each facial image is converted into a 3-channel Red, Green, Blue (RGB) format, resized to a resolution of 224×224 pixels, and its pixel intensity is scaled from the range $[0, 255]$ to $[0, 1]$ for the transformer models.

B. VISION TRANSFORMER

The visual representation extraction architecture of this study adopts ViT-Base, as illustrated in Figure 3 [26]. An input facial image of 224×224 pixels with 3 color channels is partitioned into 196 non-overlapping spatial patches $\mathbf{x}_p^i$ of size 16×16 pixels [27]. Each patch is linearly projected via matrix $\mathbf{E}$ into a feature representation space of dimension $D = 768$, and subsequently concatenated with a class token $\mathbf{x}_{\text{class}}$ and positional embedding $\mathbf{E}_{\text{pos}}$ according to (1) [27]. The initial token sequence vector $\mathbf{z}_0$ is fed into a stack of 12 identical transformer encoder layers. Each encoder layer ℓ processes the input $\mathbf{z}_{\ell-1}$ into an intermediate representation $\mathbf{z}'_\ell$ via the Multi-Head Self-Attention (MHSA) mechanism following a Layer Normalization (LN) operation according to (2), which then proceeds to an output representation $\mathbf{z}_\ell$ through a two-layer Multi-Layer Perceptron (MLP) block with GeLU activation based on (3) [28]. The self-attention mechanism enables the modeling of global relationships across facial patches without dependence on local receptive fields as in convolutional operations [29].

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \dots; \mathbf{x}_p^{196} \mathbf{E}] + \mathbf{E}_{\text{pos}} \quad (1)$$

$$\mathbf{z}'_\ell = \text{MHSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1} \quad (2)$$



FIGURE 2. Sample Images of the DemogPairs Dataset across Six Intersectional Demographic Subgroups: (a) Asian Females, (b) Asian Males, (c) Black Females, (d) Black Males, (e) White Females, and (f) White Males.

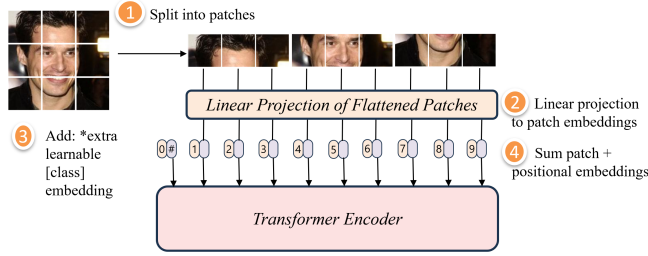


FIGURE 3. Architecture of the ViT Backbone and Patch Projection.

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell \quad (3)$$

TABLE 2. Multi-Domain Feature Fusion and Ablation Configurations.

#	Configuration	Domain Category	Dimension
1	Face	Single-Domain	768
2	Emotion	Single-Domain	768
3	Age	Single-Domain	768
4	Emotion $\oplus$ Face	Dual-Domain	1,536
5	Face $\oplus$ Age	Dual-Domain	1,536
6	Emotion $\oplus$ Age	Dual-Domain	1,536
7	Face $\oplus$ Emotion $\oplus$ Age	Tri-Domain (Proposed)	2,304

To maintain computational efficiency and prevent retraining variability, the three ViT backbone models are frozen as providers of task-associated representations via offline extraction. The ViT-Face model (skutaada/VIT-VGGFace) captures representations associated with facial biometric geometry, ViT-Emotion (dima806/facial_emotions_image_detection) produces representations associated with facial affective expression, and ViT-Age (dima806/facial_age_image_detection) extracts representations associated with facial age estimation. From each domain model, the 768-dimensional visual representation feature vector $\mathbf{f}_{\text{domain}} \in \mathbb{R}^{768}$ is extracted from the [CLS] token representation at the final encoder layer L ($\mathbf{z}_L^0$) following the LN operation according to (4). These three single-domain feature representations are combined into a tri-domain fusion vector $\mathbf{z}_{\text{tri}} \in \mathbb{R}^{2304}$ via a feature concatenation operation ($\oplus$) of the vectors $\mathbf{f}_{\text{face}}$, $\mathbf{f}_{\text{emotion}}$, and $\mathbf{f}_{\text{age}}$ as formulated in (5). The systematic exploration encompasses seven feature ablation schemes, comprising three 768-dimensional single-domain

configurations, three 1,536-dimensional dual-domain configurations, and one 2,304-dimensional tri-domain configuration, as summarized in Table 2.

$$\mathbf{f}_{\text{domain}} = \text{LN}(\mathbf{z}_L^0) \in \mathbb{R}^{768} \quad (4)$$

$$\mathbf{z}_{\text{tri}} = \mathbf{f}_{\text{face}} \oplus \mathbf{f}_{\text{emotion}} \oplus \mathbf{f}_{\text{age}} \in \mathbb{R}^{2304} \quad (5)$$

C. RANDOM FOREST

The RF ensemble model aggregates predictions from a collection of independent decision trees to classify facial features based on the principles of bootstrap aggregating (bagging) and the random subspace method [30]. This approach reduces prediction variance by constructing B classification trees from random samples of the training data [31]. Each node split restricts feature selection to a random subset of size $\text{max_features} \in \{\text{'sqrt'}, \text{'log2'}\}$ to suppress inter-tree correlation [32]. The split criterion optimizes the Gini impurity index until satisfying the minimum node split threshold min_samples_split , the maximum depth limit max_depth , or the minimum leaf threshold min_samples_leaf [33]. A majority voting mechanism determines the final decision for an image sample across all decision trees [34]. The hyperparameter tuning procedure evaluates combinations of feature scaling, PCA dimensionality reduction, tree count variations $n_{\text{estimators}}$, and tree growth constraints, producing 288 grid configurations with 1,440 total fits via 5-Fold Stratified Cross-Validation, as summarized in Table 3.

D. GAUSSIAN NAIVE BAYES

Based on Bayes' theorem, the GNB classifier estimates the posterior probabilities of intersectional demographic classes under the assumption that all continuous feature dimensions are conditionally independent [35]. The normal distribution probability density function models the continuous feature likelihood given the target class according to the formulation in (6), where x_i denotes the value of the i -th continuous feature, y is the class label variable, c represents the target class category, and $\mu_{c,i}$ and $\sigma_{c,i}^2$ denote the mean and variance of the i -th feature in class c , respectively [35]:

$$P(x_i | y = c) = \frac{1}{\sqrt{2\pi\sigma_{c,i}^2}} \exp\left(-\frac{(x_i - \mu_{c,i})^2}{2\sigma_{c,i}^2}\right) \quad (6)$$

TABLE 3. Hyperparameter Search Space for Random Forest Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Number of Estimators	100, 200	2
Maximum Tree Depth	None, 20, 30	3
Feature Subspace	'sqrt', 'log2'	2
Minimum Samples Split	2, 5	2
Minimum Samples Leaf	1, 2	2
Total Grid Combinations		288 (1,440 fits)

TABLE 4. Hyperparameter Search Space for Gaussian Naive Bayes Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Variance Smoothing	logspace($-9, 2, 40$) (1.0×10^{-9} to 1.0×10^2)	40
Total Grid Combinations		240 (1,200 fits)

The addition of the smoothing parameter `var_smoothing` maintains computational stability against the risk of near-zero variance in high-dimensional feature representation spaces [36]. The hyperparameter tuning procedure via 5-Fold Stratified Cross-Validation evaluates combinations of data scaling, PCA dimensionality reduction, and 40 `var_smoothing` intervals explored logarithmically from 1.0×10^{-9} to 1.0×10^2 , yielding 240 grid combinations with a total of 1,200 fitting processes as presented in Table 4.

E. LOGISTIC REGRESSION

The multinomial regression approach (Softmax regression) in LR maps feature representations into six intersectional demographic categories through unified linear probability estimation [37]. The Softmax probability function calculates the conditional probability of a sample belonging to the target category as defined in (7), where $\mathbf{x}$ represents the input feature vector, y is the class label variable, c denotes the target class category, $\mathbf{w}_c$ and b_c are the weight vector and bias of class c , respectively, while the constant $K = 6$ indicates the total number of intersectional classes [38]:

$$P(y = c | \mathbf{x}) = \frac{e^{\mathbf{w}_c^T \mathbf{x} + b_c}}{\sum_{j=1}^K e^{\mathbf{w}_j^T \mathbf{x} + b_j}} \quad (7)$$

The model training process minimizes the cross-entropy loss function by incorporating an L_2 regularization penalty to control weight complexity in the high-dimensional representation space [39]. The structured hyperparameter search evaluates variations in regularization strength C , optimization solver algorithms (lbfgs, saga, newton-cg), maximum iteration limits `max_iter`, as well as the integration of scaling and PCA, producing 270 evaluation combinations or 1,350 total fits under the 5-Fold Stratified Cross-Validation protocol presented in Table 5.

F. SUPPORT VECTOR MACHINE

The SVM classifier constructs a maximum margin hyperplane to separate the six intersectional classes in the high-dimensional representation space [40]. Kernel functions project the nonlinear data distribution from multi-domain fusion into a higher-dimensional feature space, encompassing linear, Radial Basis Function (RBF), and polynomial kernel options [41]. The second-degree polynomial kernel formulation in (8) computes the dot product $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ between pairs of feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, where γ denotes the kernel scaling parameter, $d = 2$ determines the mapping degree, and the intercept constant `coef0` = 0.0 follows the scikit-learn default value [42]:

$$K(\mathbf{x}_i, \mathbf{x}_j) = (\gamma \langle \mathbf{x}_i, \mathbf{x}_j \rangle + \text{coef0})^d, \quad d = 2 \quad (8)$$

This quadratic mapping facilitates the efficient capture of nonlinear interactions across visual dimensions [43].

Hyperparameter optimization evaluates penalty parameter ranges $C \in \{0.01, 0.1, 1, 10\}$, various kernel functions, γ coefficients, polynomial degrees, as well as scaling and PCA options, yielding 288 configuration combinations or 1,440 total fits across the 5-Fold Stratified Cross-Validation procedure as presented in Table 6.

G. CLASSIFICATION PIPELINE

The modular pipeline framework unifies feature scaling, representation dimensionality reduction, and downstream model estimation into an integrated transformation chain. This processing chain is formulated in (9), where the input feature vector $\mathbf{x}$ is transformed by the scaling module into $\tilde{\mathbf{x}}$, reduced in dimensionality through PCA to $\hat{\mathbf{x}}$, and mapped by the classifier to the predicted label $\hat{y} \in \{0, 1, 2, 3, 4, 5\}$ for the six demographic subgroups:

$$\mathbf{x} \xrightarrow{\text{Scaler}} \tilde{\mathbf{x}} \xrightarrow{\text{PCA}} \hat{\mathbf{x}} \xrightarrow{\text{Classifier}} \hat{y} \in \{0, 1, 2, 3, 4, 5\} \quad (9)$$

TABLE 5. Hyperparameter Search Space for Logistic Regression Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Regularization Strength	0.01, 0.1, 1, 10, 100	5
Optimization Solver	'lbfgs', 'saga', 'newton-cg'	3
Maximum Iterations	500, 1000, 2000	3
Total Grid Combinations		270 (1,350 fits)

TABLE 6. Hyperparameter Search Space for Support Vector Machine.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Regularization Parameter	0.01, 0.1, 1, 10	4
Kernel Function	'linear', 'rbf', 'poly'	3
Polynomial Degree	2, 3	2
Kernel Coefficient	'scale', 'auto'	2
Total Grid Combinations		288 (1,440 fits)

The cross-validation protocol implements a 5-Fold Stratified Cross-Validation scheme with controlled shuffling to maintain a balanced proportion of $\sim 16.67\%$ per class in each fold. To prevent information leakage, the Scaler and PCA are fitted exclusively on the training data within GridSearchCV before being applied to the validation data. Searching across 1,086 hyperparameter combinations yields 5,430 fits per feature scheme or 38,010 total fits for the seven representation configurations, along with 28 final refits prior to testing on the independent held-out test data.

H. EVALUATION METRICS

The evaluation of multi-class classification performance across the six demographic subgroups at the subgroup level is conducted via a binary One-vs-Rest (OvR) scheme [44]. This scheme utilizes four per-class confusion matrix components to calculate the OvR Accuracy, Precision, Recall, and F1-Score for each class $c \in \{1, \dots, K\}$ [45]. These four components comprise TP_c (True Positive), TN_c (True Negative), FP_c (False Positive), and FN_c (False Negative) [46]. OvR Accuracy in (10) measures the ratio of total correct predictions relative to the entire test dataset, whereas Precision in (11) and Recall in (12) map the positive predictive precision and system detection sensitivity per subgroup, respectively [47]. F1-Score in (13) is formulated as the harmonic mean between Precision and Recall to represent a balanced evaluation of detection performance per subgroup [47].

$$\text{Accuracy}_c = \frac{TP_c + TN_c}{TP_c + TN_c + FP_c + FN_c} \quad (10)$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \quad (11)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad (12)$$

$$\text{F1-Score}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (13)$$

To evaluate the overall effectiveness of the classification system on independent test data, the per-subgroup performance metrics are aggregated into global evaluation metrics. In this formulation, the symbol N denotes the total number of test samples ($N = 2,160$), whereas K represents the total number of demographic subgroups ($K = 6$). Global Accuracy (Overall Accuracy) in (14) is calculated as the ratio of total correct predictions, namely the summation of the main diagonal elements of the confusion matrix $\sum_{c=1}^K TP_c$, to the total test samples N .

$$\text{Accuracy}_{\text{Global}} = \frac{\sum_{c=1}^K TP_c}{N} \quad (14)$$

Subsequently, global Precision (Macro Precision) in (15), global Recall (Macro Recall) in (16), and global F1-Score (Macro F1-Score) in (17) are computed via an unweighted macro average across the $K = 6$ subgroups to assign equal evaluation weight to each demographic class.

$$\text{Precision}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{Precision}_c \quad (15)$$

$$\text{Recall}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{Recall}_c \quad (16)$$

$$\text{F1-Score}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{F1-Score}_c \quad (17)$$

IV. RESULTS AND DISCUSSION

A. GLOBAL PERFORMANCE ANALYSIS

Performance evaluation of RF across seven feature configurations on the DemogPairs test set is presented in Table 7. Empirical experimental results indicate that the dual-domain

TABLE 7. Performance Benchmark of Random Forest across Seven Feature Configurations.

Configuration	Accuracy	95% CI	CV Score	Precision	Recall	F1-Score	Best Parameters
Face	85.46%	[83.91%, 86.89%]	87.07% $\pm$ 0.68%	85.43%	85.46%	85.39%	n_est=200, depth=30, max_feat=log2, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=MinMaxScaler
Emotion	80.60%	[78.88%, 82.21%]	79.65% $\pm$ 1.01%	80.63%	80.60%	80.57%	n_est=200, depth=None, max_feat=log2, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None
Age	73.66%	[71.76%, 75.47%]	72.93% $\pm$ 0.85%	73.63%	73.66%	73.54%	n_est=200, depth=30, max_feat=log2, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=None
Emotion $\oplus$ Face	86.85%	[85.36%, 88.21%]	87.72% $\pm$ 0.69%	86.89%	86.85%	86.82%	n_est=200, depth=None, max_feat=sqrt, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None
Face $\oplus$ Age	85.79%	[84.25%, 87.20%]	87.20% $\pm$ 0.84%	85.78%	85.79%	85.73%	n_est=200, depth=None, max_feat=sqrt, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=None
Emotion $\oplus$ Age	81.11%	[79.41%, 82.71%]	80.07% $\pm$ 0.87%	81.11%	81.11%	81.08%	n_est=200, depth=None, max_feat=log2, min_split=5, min_leaf=2, pca=PCA(0.75), scaler=None
Face $\oplus$ Emotion $\oplus$ Age	86.20%	[84.68%, 87.59%]	87.48% $\pm$ 0.86%	86.20%	86.20%	86.13%	n_est=200, depth=30, max_feat=sqrt, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None

Emotion $\oplus$ Face scheme achieves the highest performance with an Accuracy of 86.85% and an F1-Score of 86.82%. Conversely, the tri-domain aggregation Face $\oplus$ Emotion $\oplus$ Age exhibits a slight performance degradation to an Accuracy of 86.20% and an F1-Score of 86.13%. This decline is interpreted as potential partitioning challenges in the 2,304-dimensional feature space via random tree splits [48]. All GridSearchCV configurations consistently select PCA 0.75 dimensionality reduction to balance data variance. In the single-domain category, the Face feature records the best performance with an Accuracy of 85.46%, surpassing the Emotion domain feature at 80.60% and the Age domain at 73.66%.

The performance evaluation of probabilistic GNB classification across seven feature configuration variations is summarized in Table 8. Although this model assumes conditional independence between features, GNB exhibits a consistent performance improvement trend from the single-domain to the multi-domain category. In the single-domain category, the Age configuration yields the lowest performance with an Accuracy of 69.63% and an F1-Score of 69.52%, whereas the Face feature achieves an Accuracy of 82.69%. Representation integration in the dual-domain category progressively improves accuracy, led by the Emotion $\oplus$ Face combination at 84.86%. Peak GNB classification performance is achieved by the tri-domain configuration Face $\oplus$ Emotion $\oplus$ Age with an Accuracy of 85.05% and an F1-Score of 85.05%. All GNB feature configurations uniformly select PCA 0.75 dimensionality reduction based on GridSearchCV tuning results.

The performance benchmark results of multinomial LR across all feature configurations are presented in Table 9. A key characteristic of the LR model is the retention of full feature representations, where all schemes (seven out of seven configurations) select no PCA reduction (pca=None)

[49]. Empirical evaluation demonstrates that the tri-domain configuration Face $\oplus$ Emotion $\oplus$ Age achieves the highest performance with an Accuracy of 92.73% and an F1-Score of 92.73% using penalty parameter $C = 0.1$ and the newton-cg solver. This tri-domain achievement surpasses the best dual-domain configuration Emotion $\oplus$ Face, which achieves an Accuracy of 92.41% and an F1-Score of 92.40%, as well as the best single-domain configuration Face with an Accuracy of 90.60%. These findings indicate the effectiveness of L_2 regularization in handling high-dimensional feature spaces without principal component compression [50].

The performance progression of SVM across all feature ablation stages is summarized in Table 10. Similar to the linear model, all SVM configurations consistently maintain the full representation space without PCA reduction (pca=None). Empirical evaluation demonstrates a tiered performance increase, starting from the best single-domain configuration Face with an Accuracy of 90.83% and an F1-Score of 90.83%. Representation integration in the dual-domain scheme Emotion $\oplus$ Face improves performance to an Accuracy of 93.29% and an F1-Score of 93.29%, before ultimately reaching the highest point in the tri-domain configuration Face $\oplus$ Emotion $\oplus$ Age with an Accuracy of 93.70% and an F1-Score of 93.69%. Regarding hyperparameters, the degree parameter with a value of 2 is active only for the poly kernel function, whereas RBF kernel-based models do not activate this parameter [51].

A synthesis of the 28 experiments across Table 7, Table 8, Table 9, and Table 10 establishes the tri-domain SVM Face $\oplus$ Emotion $\oplus$ Age as the best-performing model with an Accuracy of 93.70% and an F1-Score of 93.69%. The average accuracy across all configurations highlights the superiority of SVM (91.47%), followed by LR (90.40%), RF (82.81%), and GNB (79.37%). The paired McNemar test results be-

TABLE 8. Performance Benchmark of Gaussian Naive Bayes across Seven Feature Configurations.

Configuration	Accuracy	95% CI	CV Score	Precision	Recall	F1-Score	Best Parameters
Face	82.69%	[81.03%, 84.22%]	84.04% $\pm$ 0.74%	82.71%	82.69%	82.58%	var_smoothing=4.1246e-02, pca=PCA(0.75), scaler=MinMaxScaler
Emotion	73.38%	[71.48%, 75.20%]	73.11% $\pm$ 1.09%	73.87%	73.38%	73.29%	var_smoothing=3.0703e-03, pca=PCA(0.75), scaler=None
Age	69.63%	[67.66%, 71.53%]	69.19% $\pm$ 0.75%	69.79%	69.63%	69.52%	var_smoothing=4.3755e-04, pca=PCA(0.75), scaler=MinMaxScaler
Emotion $\oplus$ Face	84.86%	[83.29%, 86.31%]	85.65% $\pm$ 0.73%	84.90%	84.86%	84.81%	var_smoothing=5.8780e-03, pca=PCA(0.75), scaler=MinMaxScaler
Face $\oplus$ Age	83.15%	[81.51%, 84.67%]	84.34% $\pm$ 0.72%	83.43%	83.15%	83.17%	var_smoothing=1.1253e-02, pca=PCA(0.75), scaler=MinMaxScaler
Emotion $\oplus$ Age	76.81%	[74.98%, 78.54%]	75.54% $\pm$ 1.13%	76.86%	76.81%	76.81%	var_smoothing=1.6037e-03, pca=PCA(0.75), scaler=MinMaxScaler
Face $\oplus$ Emotion $\oplus$ Age	85.05%	[83.48%, 86.49%]	85.53% $\pm$ 0.57%	85.12%	85.05%	85.05%	var_smoothing=5.8780e-03, pca=PCA(0.75), scaler=None

TABLE 9. Performance Benchmark of Logistic Regression across Seven Feature Configurations.

Configuration	Accuracy	95% CI	CV Score	Precision	Recall	F1-Score	Best Parameters
Face	90.60%	[89.30%, 91.76%]	90.52% $\pm$ 0.59%	90.60%	90.60%	90.59%	$C = 1$, solver=newton-cg, max_iter=500, pca=None, scaler=MinMaxScaler
Emotion	88.47%	[87.06%, 89.75%]	87.18% $\pm$ 0.76%	88.50%	88.47%	88.46%	$C = 1$, solver=saga, max_iter=500, pca=None, scaler=MinMaxScaler
Age	86.48%	[84.97%, 87.86%]	84.91% $\pm$ 0.58%	86.49%	86.48%	86.48%	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Emotion $\oplus$ Face	92.41%	[91.21%, 93.45%]	91.89% $\pm$ 0.54%	92.41%	92.41%	92.40%	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Face $\oplus$ Age	91.62%	[90.38%, 92.72%]	91.53% $\pm$ 0.47%	91.62%	91.62%	91.62%	$C = 0.1$, solver=newton-cg, max_iter=500, pca=None, scaler=None
Emotion $\oplus$ Age	90.51%	[89.20%, 91.67%]	89.26% $\pm$ 0.69%	90.52%	90.51%	90.51%	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Face $\oplus$ Emotion $\oplus$ Age	92.73%	[91.56%, 93.75%]	92.19% $\pm$ 0.45%	92.75%	92.73%	92.73%	$C = 0.1$, solver=newton-cg, max_iter=500, pca=None, scaler=None

tween Tri-Domain vs. Dual-Domain for SVM record a difference of +9 images (+0.41%, $\chi^2 = 1.0492$, $p = 0.3057$, $p > 0.05$), whereas for RF a difference of -14 images is observed (-0.65%, $\chi^2 = 1.4825$, $p = 0.2232$, $p > 0.05$). The absence of statistical significance indicates the phenomenon of asymptotic feature saturation, wherein the Dual-Domain Emotion $\oplus$ Face fusion has already captured the majority of the primary discriminative variance.

B. FEATURE ABLATION STUDY

The quantification of classification performance from single-domain toward dual- and tri-domain configurations is summarized in Table 7, Table 8, Table 9, and Table 10. In the SVM model, representation transition from single-domain Face (90.83%) to dual-domain Emotion $\oplus$ Face (93.29%) increases accuracy by 2.46%. Adding a third domain in the tri-domain Face $\oplus$ Emotion $\oplus$ Age configuration further elevates accuracy to 93.70% (cumulative +2.87%), with its 95% CI [92.60%, 94.65%] overlapping the dual-domain interval [92.15%, 94.27%]. The close consistency of the CV Score (92.65% $\pm$ 0.93%) with test accuracy confirms decision boundary stability without overfitting. This progressive improvement demonstrates that combining complementary visual representations reliably sharpens intersectional demographic separation.

An analysis of the age domain (Age) examines the role of aging signals within the demographic classification system. Across all classifiers, the Age feature yields the lowest accuracy as a single domain, recording 87.64% for SVM and 69.63% for GNB. The performance decline in Random Forest (-0.65% from dual-domain to tri-domain) occurs because expanded feature dimensions make random trees prone to choosing less informative age features at split points. Conversely, the SVM model with a polynomial kernel simultaneously evaluates interactive feature dimensions. This finding refutes the premise that adding features unconditionally improves accuracy; rather, it depends on classifier characteristics in handling feature spaces, confirming that feature representations reach an empirical saturation point.

C. INTERSECTIONAL SUBGROUP PERFORMANCE

The granular classification performance profile of the best-performing tri-domain SVM model (Face $\oplus$ Emotion $\oplus$ Age) is evaluated based on the metrics in Table 11. All subgroups achieve an F1-Score exceeding 91.00%, ranging from 91.74% for Black Females to 96.14% for White Males. Nevertheless, formal fairness analysis identifies an Equal Opportunity Difference gap ($\Delta\text{TPR} = 7.50\%$) resulting from the recall disparity between White Males (96.94%) and Black Females (89.44% with an FNR of 10.56%). Although the precision for

TABLE 10. Performance Benchmark of Support Vector Machine across Seven Feature Configurations.

Configuration	Accuracy	95% CI	CV Score	Precision	Recall	F1-Score	Best Parameters
Face	90.83%	[89.54%, 91.98%]	91.41% $\pm$ 0.85%	90.84%	90.83%	90.83%	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Emotion	90.19%	[88.86%, 91.37%]	88.38% $\pm$ 0.54%	90.20%	90.19%	90.17%	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Age	87.64%	[86.18%, 88.96%]	86.78% $\pm$ 0.70%	87.67%	87.64%	87.65%	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Emotion $\oplus$ Face	93.29%	[92.15%, 94.27%]	92.31% $\pm$ 0.80%	93.33%	93.29%	93.29%	$C = 10$, rbf, γ =scale, pca=None, scaler=MinMaxScaler
Face $\oplus$ Age	92.55%	[91.36%, 93.58%]	92.30% $\pm$ 0.57%	92.54%	92.55%	92.54%	$C = 10$, poly, γ =scale, deg=2, pca=None, scaler=None
Emotion $\oplus$ Age	92.08%	[90.87%, 93.15%]	90.17% $\pm$ 0.65%	92.10%	92.08%	92.09%	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Face $\oplus$ Emotion $\oplus$ Age	93.70%	[92.60%, 94.65%]	92.65% $\pm$ 0.93%	93.72%	93.70%	93.69%	$C = 10$, poly, γ =scale, deg=2, pca=None, scaler=None

TABLE 11. Subgroup-Level Performance for Tri-Domain Models.

Classifier	Subgroup	Recall (TPR)	Precision	F1-Score	OvR Accuracy	FNR
SVM (Tri)	White_Males	96.94%	95.36%	96.14%	98.70%	3.06%
	Black_Males	94.17%	95.49%	94.83%	98.29%	5.83%
	White_Females	94.72%	92.41%	93.55%	97.82%	5.28%
	Asian_Males	94.44%	92.39%	93.41%	97.78%	5.56%
	Asian_Females	92.50%	92.50%	92.50%	97.50%	7.50%
	Black_Females	89.44%	94.15%	91.74%	97.31%	10.56%
LR (Tri)	White_Males	96.11%	95.05%	95.58%	98.52%	3.89%
	Black_Males	93.06%	95.71%	94.37%	98.15%	6.94%
	Asian_Males	92.78%	90.76%	91.76%	97.22%	7.22%
	White_Females	92.22%	91.21%	91.71%	97.22%	7.78%
	Black_Females	91.11%	92.13%	91.62%	97.22%	8.89%
	Asian_Females	91.11%	91.62%	91.36%	97.13%	8.89%

Black Females remains high at 94.15%, the lower detection sensitivity reflects intersectional phenotypic representation challenges that have not been entirely eliminated. Reporting this False Negative Rate metric balances the evaluation of OvR Accuracy (97.31% to 98.70%), which is prone to distortion by the 5:1 negative sample ratio dominance.

A comparison of subgroup performance profiles between the SVM model and the linear LR classifier under the tri-domain configuration is presented in Table 11. The tri-domain LR model exhibits high performance consistency, with F1-Scores exceeding 91.00% across all demographic classes, ranging from 91.36% for Asian Females to 95.58% for White Males. Formal fairness evaluation of the LR model demonstrates a narrower Equal Opportunity Difference gap ($\Delta\text{TPR} = 5.00\%$), with White Males achieving a recall of 96.11% compared to 91.11% for Asian Females and Black Females (FNR of 8.89%). The White Males subgroup consistently records the highest classification performance across both classifiers. These empirical findings demonstrate that tri-domain representation fusion yields stable visual representations across all demographic subgroups in both linear and non-linear models.

D. ERROR PATTERN ASSESSMENT

The shift in classification error distributions from the single-domain scheme toward dual- and tri-domain configurations is

analyzed through the confusion matrix diagrams in Figure 4, which encompass Figure 4(a), Figure 4(b), and Figure 4(c). Empirical evaluation reveals that the total misclassifications across the 2,160 held-out test samples consistently decrease as the visual representation domains expand. The single-domain Face SVM model yields 198 prediction errors, which subsequently decrease to 145 samples in the dual-domain Emotion $\oplus$ Face scheme. The tri-domain Face $\oplus$ Emotion $\oplus$ Age configuration suppresses classification errors to its lowest point of 136 samples. This consistent reduction in errors indicates that multi-domain representation fusion sharpens the separation boundaries between demographic subgroups in the visual feature space.

An in-depth examination of the confusion matrix for the single-domain Face model in Figure 4(a) reveals that classification errors are predominantly concentrated in inter-racial subgroups within the same gender. This tendency is particularly pronounced in the female cohort, where the Asian Females subgroup incurs 46 misclassifications, encompassing 16 samples predicted as White Females and 16 samples as Black Females. Conversely, the White Females subgroup records 20 prediction errors distributed to Asian Females. This empirical pattern is consistent with the potential presence of inter-racial phenotypic overlap within single facial biometric geometric representations [18]. Nevertheless, this confusion matrix evidence does not prove that visual overlap

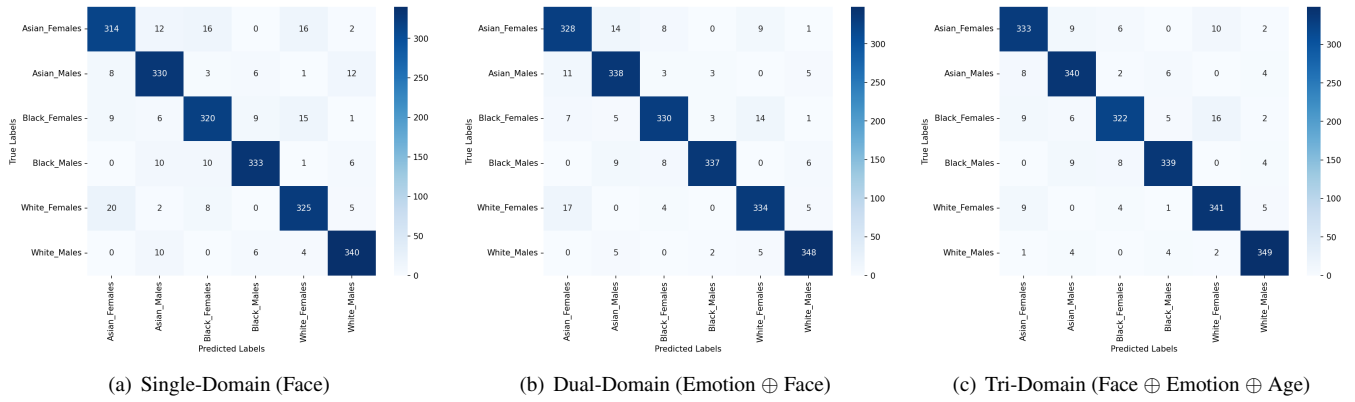


FIGURE 4. Confusion Matrices across Feature Fusion Schemes on the Held-Out Test Set: (a) Single-Domain (Face), (b) Dual-Domain (Emotion $\oplus$ Face), and (c) Tri-Domain (Face $\oplus$ Emotion $\oplus$ Age).

is the sole cause of all observed misclassifications.

The shift in classification error distributions in the dual-domain Emotion $\oplus$ Face model is further analyzed based on the confusion matrix in Figure 4(b). The addition of affective expression representations effectively reduces the inter-racial misclassification pattern among same-gender subgroups, particularly within the female group. Prediction errors from Asian Females to White Females are substantially suppressed from 16 cases to 9 cases. Concurrently, the number of correctly predicted samples (true positives) in the White Females subgroup increases from 325 images to 334 images. The integration of these dynamic facial expression signals strengthens inter-racial visual discriminant boundaries within the unified representation space, enabling the classifier to distinguish similar morphological features with a more robust and stable certainty.

The misclassification patterns in the tri-domain Face $\oplus$ Emotion $\oplus$ Age configuration are comprehensively evaluated based on the confusion matrix in Figure 4(c). The integration of age-domain representations consistently increases the number of correct predictions in the Asian Females subgroup to 333 samples and White Females to 341 samples. Furthermore, cross-gender misclassifications, representing instances where female subjects are confused as male or vice versa, amount to only 51 cases out of the total 2,160 held-out test samples (2.36%). These empirical findings indicate that inter-gender boundaries are separated very distinctly within the multi-domain representation space, such that the remaining classification errors in the tri-domain model are almost entirely dominated by inter-racial misclassifications within the same gender group.

E. COMPARISON WITH PRIOR STUDIES

A direct empirical comparison between the proposed framework and prior studies on the DemogPairs dataset is summarized in Table 12. The proposed Tri-Domain ViT + SVM model (Face $\oplus$ Emotion $\oplus$ Age) achieves an Accuracy of 93.70%, Precision of 93.72%, Recall of 93.70%, and F1-Score of 93.69%. These quantitative achievements surpass

TABLE 12. Comparative Performance of Proposed Framework against Prior Studies on the DemogPairs Dataset.

Model	Accuracy	Precision	Recall	F1-Score
MD-ViT [20]	89.07%	89.12%	89.07%	89.01%
Dual-ViT [19]	92.41%	92.48%	92.41%	92.38%
Ours	93.70%	93.72%	93.70%	93.69%

the performance reported for the Dual-ViT + SVM model with 92.41% Accuracy and 92.38% F1-Score [19], as well as the MD-ViT + XGBoost model with 89.07% Accuracy and 89.01% F1-Score [20]. This comparison positions the scientific contribution of this paper as a unified systematic tri-domain evaluation featuring significance testing and decision boundary analyses that extend beyond prior conference proceedings studies [19]. All comparative figures are cited directly from their respective publications as a contextual positioning on the same reference dataset.

The seminal research introducing the DemogPairs dataset by Hupont and Fernández [25] was exclusively designed as an identity verification benchmark (identity verification ROC across 58.3 million image pairs) and did not perform 6-class classification training (closed-set intersectional classification). Therefore, the present study, together with related prior work [19], [20], pioneers and provides the first comprehensive internally controlled baselines for 6-class intersectional demographic classification on the DemogPairs dataset. In addition to evaluating global aggregate metrics, this investigation conducts an in-depth subgroup disparity analysis on the best tri-domain configuration to transparently map performance variations across demographic subgroups. This thorough evaluation demonstrates classification stability across intersectional race and gender categories, whereas subgroup performance variation information was not reported in prior comparative literature [19], [20].

V. CONCLUSION

This study evaluates a Tri-Domain ViT representation fusion framework (Face $\oplus$ Emotion $\oplus$ Age: 2,304 dimensions) for intersectional race and gender classification on the DemogPairs dataset. Experimental results demonstrate that tri-domain fusion serves as the best configuration across 3 of the 4 evaluated classifiers, encompassing SVM, LR, and GNB. Within the evaluated search space, the optimized SVM model with hyperparameter $C = 10$ and a second-degree polynomial kernel yields the highest classification performance, namely an Accuracy of 93.70% and an F1-Score of 93.69%. Nevertheless, the benefit of multi-domain representation integration proves to be classifier-dependent, as RF achieves its peak performance on the dual-domain Emotion $\oplus$ Face scheme with an accuracy of 86.85%. In addition, subgroup performance evaluation confirms that the F1-Score values across all six demographic subgroups are stably distributed within the range of 91.74% to 96.14%.

Despite these achievements, this study has several limitations that open avenues for future research. First, the applied dataset partitioning scheme represents a deliberate limitation, wherein the in-distribution benchmark protocol focuses on evaluating isolated representations at the image level. Second, the demographic scope of this investigation is limited to three macro-racial groups (Asian, Black, and White) in the DemogPairs dataset; thus, the empirical findings do not yet comprehensively cover multiracial populations or other ethnic groups. Third, feature representation extraction is conducted separately (offline) utilizing frozen pre-trained ViT backbones without simultaneous end-to-end fine-tuning. Fourth, the experimental evaluation remains focused on controlled laboratory environment images, such that model robustness against extreme pose variations, occlusions, and in-the-wild illumination requires further investigation. Based on these limitations, future research directions are recommended to formulate cross-identity generalization testing through strictly subject-disjoint partitioning that ensures the same subjects do not appear simultaneously in both training and test sets. Furthermore, future work should explore cross-modal adaptive attention fusion mechanisms (cross-attention transformer fusion) trained end-to-end, extend evaluation to large-scale heterogeneous datasets such as FairFace or UTK-Face, and integrate visual explainability methods such as Attention Rollout or Grad-CAM to uphold artificial intelligence accountability principles with human-in-the-loop oversight.

DATA AVAILABILITY STATEMENT

The DemogPairs dataset analyzed in this study is a publicly available secondary dataset for academic research purposes. Facial images along with demographic label annotations can be freely accessed via the official developer publication at <https://ihupont.github.io/publications/2019-05-16-demogpairs> and the official Google Drive repository at https://drive.google.com/file/d/1f_ez-ll6wxDXScrG4ceStZRuLT_K8Uy8/view?usp=sharing.

REFERENCES

- [1] M. K. Rusia and D. K. Singh, "A comprehensive survey on techniques to handle face identity threats: challenges and opportunities," *Multimedia Tools and Applications*, vol. 82, no. 2, pp. 1669–1748, Jan. 2023.
- [2] B. Meden, P. Rot, P. Terhöst, N. Damer, A. Kuijper, W. J. Scheirer, A. Ross, P. Peer, and V. Štruc, "Privacy-enhancing face biometrics: a comprehensive survey," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 4147–4183, 2021.
- [3] A. Birhane, "The unseen Black faces of AI algorithms," *Nature*, vol. 610, no. 7932, pp. 451–452, Oct. 2022.
- [4] H. Overbye-Thompson, K. A. Hamilton, and D. Mastro, "Reinvention mediates impacts of skin tone bias in algorithms: implications for technology diffusion," *Journal of Computer-Mediated Communication*, vol. 29, no. 5, p. zmae016, Sept. 2024.
- [5] P. Terhöst, J. N. Kolf, M. Huber, F. Kirchbuchner, N. Damer, A. M. Moreno, J. Fierrez, and A. Kuijper, "A comprehensive study on face recognition biases beyond demographics," *IEEE Transactions on Technology and Society*, vol. 3, no. 1, pp. 16–30, Mar. 2022.
- [6] N. Liu, F. Zhang, L. Chang, and F. Duan, "Facial attribute classification by deep mining inter-attribute correlations," *IET Computer Vision*, vol. 17, no. 3, pp. 352–365, 2023.
- [7] D. R. Yoder-Himes, A. Asif, K. Kinney, T. J. Brandt, R. E. Cecil, P. R. Himes, C. Cashon, R. M. P. Hopp, and E. Ross, "Racial, skin tone, and sex disparities in automated proctoring software," *Frontiers in Education*, vol. 7, p. 881449, 2022.
- [8] Z. S. Marzoog, A. D. Hasan, and H. H. Abbas, "Gender and race classification using geodesic distance measurement," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 27, no. 2, pp. 820–831, Aug. 2022.
- [9] J. P. Robinson, C. Qin, Y. Henon, S. Timoner, and Y. Fu, "Balancing biases and preserving privacy on Balanced Faces in the Wild," *IEEE Transactions on Image Processing*, vol. 32, pp. 4365–4377, Aug. 2023.
- [10] M. Sohail, G. Ali, J. Rashid, I. Ahmad, S. H. Almotiri, M. A. AlGhamdi, A. A. Nagra, and K. Masood, "Racial identity-aware facial expression recognition using deep convolutional neural networks," *Applied Sciences*, vol. 12, no. 1, p. 88, 2022.
- [11] D. Belcar, P. Grd, and I. Tomićić, "Automatic ethnicity classification from middle part of the face using convolutional neural networks," *Informatics*, vol. 9, no. 1, p. 18, 2022.
- [12] H. Liao, L. Yuan, M. Wu, L. Zhong, G. Jin, and N. Xiong, "Face gender and age classification based on multi-task, multi-instance and multi-scale learning," *Applied Sciences*, vol. 12, no. 23, p. 12432, 2022.
- [13] G. Sunitha, K. Geetha, S. Neelakandan, A. K. S. Pundir, S. Hemalatha, and V. Kumar, "Intelligent deep learning based ethnicity recognition and classification using facial images," *Image and Vision Computing*, vol. 121, p. 104404, 2022.
- [14] J. Brinkmann, P. Swoboda, and C. Bartelt, "A multidimensional analysis of social biases in vision transformers," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 4891–4900.
- [15] S. Ramachandran and A. Rattani, "Deep generative views to mitigate gender classification bias across gender-race groups," in *Pattern Recognition, Computer Vision, and Image Processing. ICPR 2022 International Workshops and Challenges*, J.-J. Rousseau and B. Kapralos, Eds. Cham, Switzerland: Springer Nature Switzerland, 2023, pp. 551–569.
- [16] S. Chen, X. Lai, Y. Yan, D.-H. Wang, and S. Zhu, "Learning an attention-aware parallel sharing network for facial attribute recognition," *Journal of Visual Communication and Image Representation*, vol. 90, p. 103745, 2023.
- [17] G. G. Tahyudin, M. D. Sulistiyo, M. Arzaki, and E. Rachmawati, "Classifying gender based on face images using vision transformer," *International Journal on Informatics Visualization*, vol. 8, no. 1, pp. 18–25, 2024.
- [18] A.-A. Kalkatawi and U. Saeed, "Ethnicity classification based on facial images using deep learning approach," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 2, pp. 217–226, 2024.
- [19] R. A. Putri, L. Anifah, R. E. Putra, Y. Yamasari, and R. A. Akbar, "Dual vision transformer integration for race and gender recognition based on facial images," in *Proceedings of the 2025 8th International Conference on Vocational Education and Electrical Engineering (ICVEE)*, 2025, pp. 258–264.
- [20] R. A. Putri, R. E. Putra, and Y. Yamasari, "MD-ViT: Multidomain vision transformer fusion for fair demographic attribute recognition," *Journal of Information Engineering and Educational Technology*, vol. 9, no. 2, pp. 64–79, 2025.

- [21] S. K. Gupta and N. Nain, "Review: single attribute and multi attribute facial gender and age estimation," *Multimedia Tools and Applications*, vol. 82, no. 1, pp. 1289–1311, 2023.
- [22] Y. Deng, S. Teng, L. Fei, W. Zhang, and I. Rida, "A multifeature learning and fusion network for facial age estimation," *Sensors*, vol. 21, no. 13, p. 4597, 2021.
- [23] Z. Wang, L. Xiao, Z. Cao, and H. Lu, "Vision transformer off-the-shelf: a surprising baseline for few-shot class-agnostic counting," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 38, no. 6, Mar. 2024, pp. 5832–5840.
- [24] D. Wilimitis and C. G. Walsh, "Practical considerations and applied examples of cross-validation for model development and evaluation in health care: tutorial," *JMIR AI*, vol. 2, p. e49023, 2023.
- [25] I. Hupont and C. Fernández, "DemogPairs: Quantifying the impact of demographic imbalance in deep face recognition," in *Proceedings of the 2019 14th IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, 2019, pp. 1–7.
- [26] K. Han, Y. Wang, H. Chen, X. Chen, J. Guo, Z. Liu, Y. Tang, A. Xiao, C. Xu, Y. Xu, Z. Yang, Y. Zhang, and D. Tao, "A survey on vision transformer," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 1, pp. 87–110, Jan. 2023.
- [27] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Computing Surveys*, vol. 54, no. 10s, pp. 200:1–200:41, Sept. 2022.
- [28] Y. Liu, Y. Zhang, Y. Wang, F. Hou, J. Yuan, J. Tian, Y. Zhang, Z. Shi, J. Fan, and Z. He, "A survey of visual transformers," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 6, pp. 7478–7498, June 2024.
- [29] J. Ma, Y. Bai, B. Zhong, W. Zhang, T. Yao, and T. Mei, "Visualizing and understanding patch interactions in vision transformer," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 10, pp. 13 671–13 680, 2024.
- [30] M. A. Ganaie, M. Tanveer, P. N. Suganthan, and V. Snasel, "Oblique and rotation double random forest," *Neural Networks*, vol. 153, pp. 496–517, 2022.
- [31] C. Zhao, D. Wu, J. Huang, Y. Yuan, H.-T. Zhang, R. Peng, and Z. Shi, "BoostTree and BoostForest for ensemble learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 7, pp. 8110–8126, 2023.
- [32] Y.-X. He, S.-H. Lyu, and Y. Jiang, "Interpreting deep forest through feature contribution and MDI feature importance," *ACM Transactions on Knowledge Discovery from Data*, vol. 20, no. 1, p. 15, Dec. 2025.
- [33] N. Zhu, C. Zhu, L. Zhou, Y. Zhu, and X. Zhang, "Optimization of the random forest hyperparameters for power industrial control systems intrusion detection using an improved grid search algorithm," *Applied Sciences*, vol. 12, no. 20, p. 10456, 2022.
- [34] J. Chen, X. Wang, and F. Lei, "Data-driven multinomial random forest: a new random forest variant with strong consistency," *Journal of Big Data*, vol. 11, no. 1, p. 34, Feb. 2024.
- [35] E. Redivo, C. Viroli, and A. Farcomeni, "Quantile-distribution functions and their use for classification, with application to naïve Bayes classifiers," *Statistics and Computing*, vol. 33, no. 2, p. 55, Mar. 2023.
- [36] Y. A. Khan, S. Imaduddin, Y. P. Singh, M. Wajid, M. Usman, and M. Abbas, "Artificial intelligence based approach for classification of human activities using MEMS sensors data," *Sensors*, vol. 23, no. 3, p. 1275, Jan. 2023.
- [37] A. J. Molstad and K. Motwani, "Multiresolution categorical regression for interpretable cell-type annotation," *Biometrics*, vol. 79, no. 4, pp. 3485–3496, Dec. 2023.
- [38] P. Zhang, R. Wang, and N. Xiu, "Multinomial logistic regression classifier via lq,0-proximal Newton algorithm," *Neurocomputing*, vol. 468, pp. 148–164, 2022.
- [39] Y. Zhang, C. Wei, and X. Liu, "Group logistic regression models with l_{p,q} regularization," *Mathematics*, vol. 10, no. 13, p. 2227, 2022.
- [40] Z. Lai, X. Chen, J. Zhang, H. Kong, and J. Wen, "Maximal margin support vector machine for feature representation and classification," *IEEE Transactions on Cybernetics*, vol. 53, no. 10, pp. 6700–6713, 2023.
- [41] C. Yang, S.-K. Oh, B. Yang, W. Pedrycz, and L. Wang, "Hybrid fuzzy multiple SVM classifier through feature fusion based on convolution neural networks and its practical applications," *Expert Systems with Applications*, vol. 202, p. 117392, 2022.
- [42] M. A. Chandra and S. S. Bedi, "Survey on SVM and their application in image classification," *International Journal of Information Technology*, vol. 13, no. 5, pp. 1–11, Oct. 2021.
- [43] R. Guido, M. C. Groccia, and D. Conforti, "A hyper-parameter tuning approach for cost-sensitive support vector machine classifiers," *Soft Computing*, vol. 27, no. 18, pp. 12 863–12 881, Sept. 2023.
- [44] H. H. Rashidi, S. Albahra, S. Robertson, N. K. Tran, and B. Hu, "Common statistical concepts in the supervised machine learning arena," *Frontiers in Oncology*, vol. 13, p. 1130229, Feb. 2023.
- [45] I. M. De Diego, A. R. Redondo, R. R. Fernández, J. Navarro, and J. M. Moguerza, "General performance score for classification problems," *Applied Intelligence*, vol. 52, no. 10, pp. 12 049–12 063, Aug. 2022.
- [46] A. Tharwat, "Classification assessment methods," *Applied Computing and Informatics*, vol. 17, no. 1, pp. 168–192, Jan. 2021.
- [47] S. A. Hicks, I. Strümke, V. Thambawita, M. Hammou, M. A. Riegler, P. Halvorsen, and S. Parasa, "On evaluation metrics for medical applications of artificial intelligence," *Scientific Reports*, vol. 12, no. 1, p. 5979, Apr. 2022.
- [48] G. Hanczár, M. Stippinger, D. Hanák, M. T. Kurbucz, O. M. Törteli, Á. Chripkó, and Z. Somogyvári, "Feature space reduction method for ultrahigh-dimensional, multiclass data: random forest-based multi-round screening (RFMS)," *Machine Learning: Science and Technology*, vol. 4, no. 4, p. 045012, Oct. 2023.
- [49] S. Guo, J. Zhang, Y. Wu, A. C. McLain, J. W. Hardin, B. Olatosi, and X. Li, "Functional multivariable logistic regression with an application to HIV viral suppression prediction," *Biometrical Journal*, vol. 66, no. 5, p. e202300081, 2024.
- [50] K. Nadeem and M.-A. Jabri, "Stable variable ranking and selection in regularized logistic regression for severely imbalanced big binary data," *PLOS ONE*, vol. 18, no. 1, pp. 1–26, Jan. 2023.
- [51] Scikit-learn Developers, "SVC," <https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html>, 2026, accessed: Feb. 2, 2026.



RICKY EKA PUTRA received the Bachelor's, Master's, and Doctoral degrees in Computer Science from Institut Teknologi Sepuluh Nopember (ITS) in 2008, 2011, and 2020, respectively, and obtained the Professional Engineer degree in 2024. Currently, he serves as a lecturer and researcher at the Department of Informatics, Faculty of Informatics, Universitas Negeri Surabaya. He is actively engaged in the implementation of the three pillars of higher education, encompassing informatics teaching, community empowerment programs, scientific publications in reputable journals, research dissemination at scientific seminars, and the acquisition of Intellectual Property Rights (IPR). His primary research focus is directed toward the development of applied artificial intelligence models that bridge advancements in intelligent computational methods with the resolution of real-world data-driven challenges. His research interests include Deep Learning, Computer Vision, Big Data Analytics, digital applications and user experience, clinical psychology, behavioral intention, healthcare informatics, statistical methods, as well as digital and early childhood education.



REZKY ARISANTI PUTRI received the Bachelor's degree in Informatics Engineering in 2019 with a specialization in Artificial Intelligence and completed her Master's degree in Informatics at Universitas Negeri Surabaya (UNESA). Currently, she actively serves as a researcher at the Department of Informatics, Faculty of Informatics, Universitas Negeri Surabaya. She is intensively involved in research on facial demographic attribute recognition based on transformer models and actively disseminates her scholarly contributions through publications in international conferences and reputable scientific journals. Her research focus and interests include Computer Vision, Deep Learning, Vision Transformers, Facial Attribute Recognition, Natural Language Processing, Machine Learning, and Algorithmic Fairness. Through consistent academic activities and research exploration, she continues to deepen her scientific competence and make tangible contributions to the advancement of intelligent computing technology.



SHUKOR SANIM MOHD FAUZI is a Senior Lecturer with the Faculty of Computer and Mathematical Sciences, Universiti Teknologi MARA (UiTM), Malaysia. He received the Diploma in Computer Science and the Bachelor of Science (Hons.) degree in Information Systems Engineering from UiTM, followed by a Master's degree from the Centre for Advanced Software Engineering, Universiti Teknologi Malaysia, and earned the Ph.D. degree in Software Engineering from the University of New South Wales (UNSW), Australia. His research interests encompass software engineering and empirical software engineering, with particular focus on mining software repositories, social network analysis, socio-technical congruence, computer-supported collaborative work, as well as software process analysis and development.

...



YUNI YAMASARI is a lecturer and researcher at the Department of Informatics, Faculty of Informatics, Universitas Negeri Surabaya (UNESA), with a solid academic foundation in both scholarly and practical domains. She received the Bachelor's, Master's, and Doctoral degrees from Institut Teknologi Sepuluh Nopember (ITS), with primary expertise in Software Engineering and Data Mining. Her activities across the three pillars of higher education are characterized by active involvement in community empowerment, publications of reputable scientific articles, and academic textbook authorship. She periodically disseminates her research findings across various national and international scientific forums and has acquired several Intellectual Property Rights (IPR). Her academic dedication is directed toward advancing software engineering, comprehensive data analysis, and applied informatics.



RAFY AULIA AKBAR received the Bachelor's degree in Informatics Engineering in 2019 and earned his Master's degree in Informatics from Universitas Negeri Surabaya with a concentration in Computer Science. During his studies, he achieved distinction by receiving the Best Solution Award at the regional competition ASEAN Vocational and Engineering Camp 2017 themed "Toward School 4.0". Currently, he actively dedicates himself as a researcher at the Department of Informatics, Faculty of Informatics, Universitas Negeri Surabaya. His research focus and interests center on the domains of Computer Vision, Medical Image Analysis, Multimodal Deep Learning, Natural Language Processing, and Vision Transformer architectures, particularly in the development of automated medical report generation systems.